Alberto Fernández-de-Marcos

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PhD Candidate · Department of Statistics · Carlos III University of Madrid · Madrid, Spain

Summary

My first research experience came during my bachelor's degree in aerospace engineering, when I worked on applications of statistical mechanics to financial data. Later, working as a data scientist in industry motivated me to deepen my understanding of statistical methodology through postgraduate studies. My interest in education led me to teach statistical learning as a part-time university lecturer and mathematics in secondary education. While teaching, I began my PhD part time in 2021. Two years later, I secured a permanent teaching position at a state secondary school through competitive examinations, ranking among the top 10% of candidates. In 2024, I was awarded a predoctoral fellowship and resigned from my teaching position to pursue my PhD full time. I am now completing it and expect to defend my thesis in January 2027.

Research

My research focuses on directional and nonparametric statistics, particularly tests of uniformity and homogeneity for circular and spherical data in arbitrary dimensions. I develop asymptotic theory and implement these methods in software, with attention to applications in other fields. My main contribution to uniformity testing is *On m-points uniformity tests on hyperspheres* (accepted in *Bernoulli*). In this paper, we introduce a general framework that extends the class of Sobolev tests through higher-order interactions, improving power in several settings. In a recent paper, *A class of nonparametric homogeneity tests on the circle*, we provide a unified, distribution-free framework for comparing multiple populations of circular data. It introduces flexible aggregation rules and the first Anderson–Darling-type circular homogeneity test. We establish consistency and derive limiting distributions under the null hypothesis and local alternatives. We also identify a resonance phenomenon arising from discrete scores that causes a classical asymptotic result to fail.

Education

2021–Jan 2027 **PhD in Mathematical Engineering**
Carlos III University of Madrid (UC3M). Community of Madrid PIPF fellowship.
Contributions on one- and two-sample tests with directional data.
Supervisor: Eduardo García-Portugués.
Part-time, 2021–2024; full-time since October 2024.

2021 **MSc in Statistics for Data Science**
Carlos III University of Madrid (UC3M).

2021 **Master's in Teacher Training for Secondary Education (Mathematics)**
Rey Juan Carlos University (URJC).

2018 **BSc in Aerospace Engineering**
Technical University of Madrid (UPM).

Awards, distinctions and fellowships

2025 **Mardia Poster Award · ADISTA 2025**
Awarded for the best poster at the International Workshop on Advances in Directional Statistics.

2024 **PIPF Predoctoral Fellowship · Community of Madrid**
Competitive four-year doctoral funding; 2023 call, awarded in 2024.
PIPF-2023/COM-30702.

2023 **Competitive Examinations · State Secondary-School Teaching**
Ranked among the top 10% of 2,038 candidates and secured a permanent mathematics teaching position in the Community of Madrid.
Resigned from the position in 2024 to pursue doctoral research full time.

Publications

Citation metrics

Web of Science
Citations: **45**
h-index: **3**

Google Scholar
Citations: **85**
h-index: **3**

Statistical methodology

1 **Fernández-de-Marcos, A.** and García-Portugués, E. (2026). A class of nonparametric homogeneity tests on the circle. Submitted to . doi:10.48550/arXiv.2609.13019

2 **Fernández-de-Marcos, A.**, García-Portugués, E., and Verdebout, T. (2026). On m-points uniformity tests on hyperspheres. *Bernoulli*, to appear. Manuscript

3 **Fernández-de-Marcos, A.** and García-Portugués, E. (2024). A stereographic test of spherical uniformity. *Statistics & Probability Letters*, 215:110218. doi:10.1016/j.spl.2024.110218

4 **Fernández-de-Marcos, A.** and García-Portugués, E. (2023). On new omnibus tests of uniformity on the hypersphere. *TEST*, 32(4):1508–1529. doi:10.1007/s11749-023-00882-x

5 **Fernández-de-Marcos, A.** and García-Portugués, E. (2023). Data-driven stabilizations of goodness-of-fit tests. *Computational Statistics & Data Analysis*, 179:107653. doi:10.1016/j.csda.2022.107653

Interdisciplinary research

6 Rodríguez-Sanz, A., **Fernández-de-Marcos, A.**, Pérez-Castán, J. A., Gómez Comendador, F., Arnaldo Valdés, R., and París Loreiro, A. (2021). Queue behavioural patterns for passengers at airport terminals: A machine learning approach. *Journal of Air Transport Management*, 90:101940. doi:10.1016/j.jairtraman.2020.101940

7 Ruiz, G. and **Fernández-de-Marcos, A.** (2018). Evidence for criticality in financial data. *The European Physical Journal B*, 91(1):article 1. doi:10.1140/epjb/e2017-80535-3

Projects

Research projects

2024–2028 **STENED · Stein-based goodness-of-fit TEsts for Non-Euclidean Data**
Team member, UC3M; joint project with Karlsruhe Institute of Technology, Germany.
Principal investigators: Eduardo García-Portugués and Bruno Ebner.
Funding: AEI (Spain) and DFG (Germany) · DFG-AEI-2023-STENED.

2022–2026 **NIDIR · Nonparametric Inference for Directional Data**
Team member, Carlos III University of Madrid.
Principal investigator: Eduardo García-Portugués.
Funding: Agencia Estatal de Investigación (AEI) · PID2021-124051NB-I00.

Other projects

2025 **EuroCC Spain · High-performance computing testbed**
Project lead, UC3M; Galicia Supercomputing Center (CESGA).
Access to high-performance computing resources: 50,000 computing hours.
Co-funded by the European Union.

Courses

2024 **HiTEc Winter Course · London**
Regularization, Bayesian nonparametrics, and robust volatility modelling.
21-hour course; supported by a HiTEc PhD grant.

Refereeing

Reviewer for: Journal of Computational and Graphical Statistics, Journal of Applied Statistics, Boletín de Estadística e Investigación Operativa.

Conference presentations

2026	1	A class of k-sample tests for circular data. XLII Congreso Nacional de Estadística, Investigación Operativa y Ciencia de Datos (SEIO 2026). Santiago de Compostela, Spain, 2–5 Sep.
	2	Testing homogeneity across circular k-samples. 7th International Symposium on Nonparametric Statistics (ISNPS 2026). Thessaloniki, Greece, 22–26 Jun.
2025	3	On m-points tests for uniformity on hyperspheres. IMS International Conference on Statistics and Data Science (ICSDS 2025). Sevilla, Spain, 15–18 Dec. Poster.
	4	Testing uniformity on the sphere: from pairs to m-tuples. Advances in Directional Statistics (ADISTA 2025). Esch-sur-Alzette, Luxembourg, 10–12 Sep. Poster. **Mardia Poster Award**.
	5	Testing uniformity on the sphere: from pairs to m-tuples. 8th International Conference on Econometrics and Statistics (EcoSta 2025). Tokyo, Japan, 21–23 Aug. **Invited talk**.
	6	Testing uniformity on the sphere: from pairs to m-tuples. Workshop on Nonparametric Statistics (NP-SEIO 2025). Mieres, Spain, 16–18 Jul.
	7	Testing uniformity on the sphere via m-points. XLI Congreso Nacional de Estadística e Investigación Operativa (SEIO 2025). Lleida, Spain, 10–13 Jun.
2024	8	An m-points class of uniformity tests on the sphere. IMS International Conference on Statistics and Data Science (ICSDS 2024). Nice, France, 16–19 Dec.
	9	On an m-points class of uniformity tests on the sphere. II Joint Workshop on Functional Data Analysis and Nonparametric Statistics (II JWFDANP). Miraflores de la Sierra, Spain, 10–13 Sep.
	10	Tests of uniformity on the sphere with data-driven parameters. 6th International Symposium on Nonparametric Statistics (ISNPS 2024). Braga, Portugal, 25–29 Jun.
2023	11	Tests of uniformity on the sphere with data-driven parameters. XL Congreso Nacional de Estadística e Investigación Operativa (SEIO 2023). Elche, Spain, 7–10 Nov.
2022	12	On new omnibus tests of uniformity on the hypersphere. 3rd Spanish Young Statisticians and Operational Researchers Meeting (SYSORM 2022). Elche, Spain, 21–23 Sep.
	13	Data-driven stabilizations of goodness-of-fit tests. Advances in Directional Statistics (ADISTA 2022). Santiago de Compostela, Spain, 15–17 Jun. Poster.

Teaching

2025–2026	**Predoctoral Teaching Assistant · UC3M** Statistics (BSc Mechanical Engineering, year 1): 6 ECTS; 30 teaching hours. Predictive Modeling (BSc Data Science and Engineering, year 2): 6 ECTS; 30 teaching hours.
2021–2024	**Secondary School Mathematics Teacher · Community of Madrid** State-subsidized schools, 2021–2023; state schools, 2023–2024. Approx. 2,500 hours taught.
2021–2022	**Part-time Lecturer (Profesor Asociado) · UC3M** Statistical Learning (BSc Management and Technology, year 4): 6 ECTS; 60 teaching hours. **Teaching distinction:** recognition for excellent student evaluations.

Industry experience

2019–2020	**Boston Consulting Group (BCG)** Planning telecommunications network deployment. Optimizing loyalty discount campaigns.
2018–2019	**Kernel Analytics** Optimizing assortments of do-it-yourself (DIY) products.
2018–2018	**Sogeti (Capgemini Group)** Predictive maintenance and reliability models for a European aircraft manufacturer.
2017–2018	**cubenube** Supervised machine learning (R). Web scraping (R, Python, SQL). Spatial data visualization (R, HTML, QGIS).